## Laboratory Information System

The laboratory information system is a sophisticated, customized computer software package used by health care organizations and laboratories to define the considerations unique to the facility. The benefits of using an LIS include quicker turnaround time, decreased clerical and billing errors, flexible delivery options for reports, and increased efficiency while saving costs in the laboratory. Programs within the laboratory information system allow phlebotomists to enter specific tasks such as entering testing orders, evaluating and entering results, inquiring about results, and generating reports. The integrity of the data entered directly impacts the reliability of test results, the efficiency of laboratory workflows, and the ability of health care providers to make informed decisions. Precise data entry ensures that specimens are correctly matched to patients, reducing the risk of misdiagnosis or inappropriate treatment. The LIS can generate ***mnemonic*** codes to identify the type and volume of a tube required to conduct a test. For example, if phlebotomists know that a complete blood count is covered, the code on the label will read 5 mL LAV. This indicates to the phlebotomists the amount required for the tube and the color of the tube. This facilitates the phlebotomist in acquiring the proper specimen and minimizing errors. The workflow for obtaining a specimen through the LIS consists of the following.

- A test request is entered into the computer.
- A collection label is printed.
- The sample is collected.
- The sample is delivered to the lab.
- The bar code is scanned (which verifies the collection and receipt).
- Orders are uploaded to an interface.
  - The nurse or provider checks the results.
- A technician verifies results.
- Test results are sent to the nursing station or accumulated in the patient record.
- The patient's laboratory charges are sent to the accounting and billing department.

### LIS-Barcoding

Barcoding is used to identify the collection and receipt of specimens, and it is considered cost-effective and used more commonly in the health care setting.

***mnemonic.*** Something that aids memory and recall.

As a phlebotomist, ensuring that laboratory results are accurately distributed to ordering providers and that the medical record is updated is a critical component of patient care. To facilitate this, a phlebotomist must diligently follow the established protocols for labeling and tracking specimens from the moment of collection to the delivery to the laboratory. Using a laboratory information system, the phlebotomist can monitor the progress of the tests and confirm that results are reported in a timely manner. Once results are available, the phlebotomist should verify that they are correctly entered into the patient's electronic health record and promptly communicated to the ordering provider. This may involve direct notification for critical values that require immediate attention. By maintaining open lines of communication with laboratory technicians and health care providers and by ensuring that all data is accurately recorded, the phlebotomist plays a vital role in the continuum of care, supporting timely clinical decision-making and contributing to positive patient outcomes.

## Reporting Critical Values (Summary)

Laboratory test outcomes and specimen data can be communicated through various methods. In laboratories equipped with full automation and computer systems, an electronic report is produced by the laboratory information system once the results are confirmed. This report is subsequently integrated with medical records and the health information system, allowing access to the physician who ordered the tests and/or other authorized individuals. In the absence of an automated reporting system, reports printed by the instruments can be delivered to the relevant department or the office of the health care provider.

***Point-of-care testing*** (POCT), also known as alternative testing, bedside, or near-patient testing, brings laboratory testing to the patient. POCTs have been made available through advancements in technology with portable handheld devices. The benefit of POCT is the accessibility to the patient and shorter times for results so the health care provider can address patient needs, deliver faster medical attention, and improve patient outcomes. Many point-of-care tests are waived test, but not all. If a POCT is not a waived test, a certified phlebotomist should not perform that test.

### Reporting Critical Values (Summary)

When conveying critical values, the phlebotomist must ensure that the individual on the receiving end is permitted to obtain the information, record the details provided, and then repeat the information back to the phlebotomist to confirm the correctness of the recorded data.

***point-of-care testing.*** The testing that is brought to the patient's location. Also known as alternate site testing (AST), ancillary, bedside, or near-patient testing.